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(19) **United States**(12) **Patent Application Publication**
KAMEZAKI et al.(10) **Pub. No.: US 2025/0277358 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **FLUSH WATER TANK DEVICE, AND FLUSH
TOILET APPARATUS INCLUDING THE
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CPC **E03D 1/34** (2013.01); **E03D 2201/30**
(2013.01)(71) Applicant: **TOTO LTD.**, Fukuoka (JP)(72) Inventors: **Mao KAMEZAKI**, Fukuoka (JP);
Hideaki KASHIMURA, Fukuoka (JP);
Haruki MATSUDA, Fukuoka (JP);
Koki SHINOHARA, Fukuoka (JP);
Mitsuhiro SUENAGA, Fukuoka (JP)(57) **ABSTRACT**

The present invention provides a flush water tank device, including a flush water tank main body including a first tank part and a second tank part for storing flush water, a rim spouting switch mechanism that switches discharge and stop of flush water in the first tank part, and a jet spouting discharge valve that switches discharge and stop of flush water in the second tank part, the first tank part and the second tank part are configured to have different cross-sectional areas depending on a height, and a ratio of the cross-sectional area of the first tank part to that of the second tank part at a first height is different from a ratio of the cross-sectional area at a second height that is lower than the first height.

(21) Appl. No.: **19/060,077**(22) Filed: **Feb. 21, 2025**(30) **Foreign Application Priority Data**

Feb. 29, 2024 (JP) 2024-029551

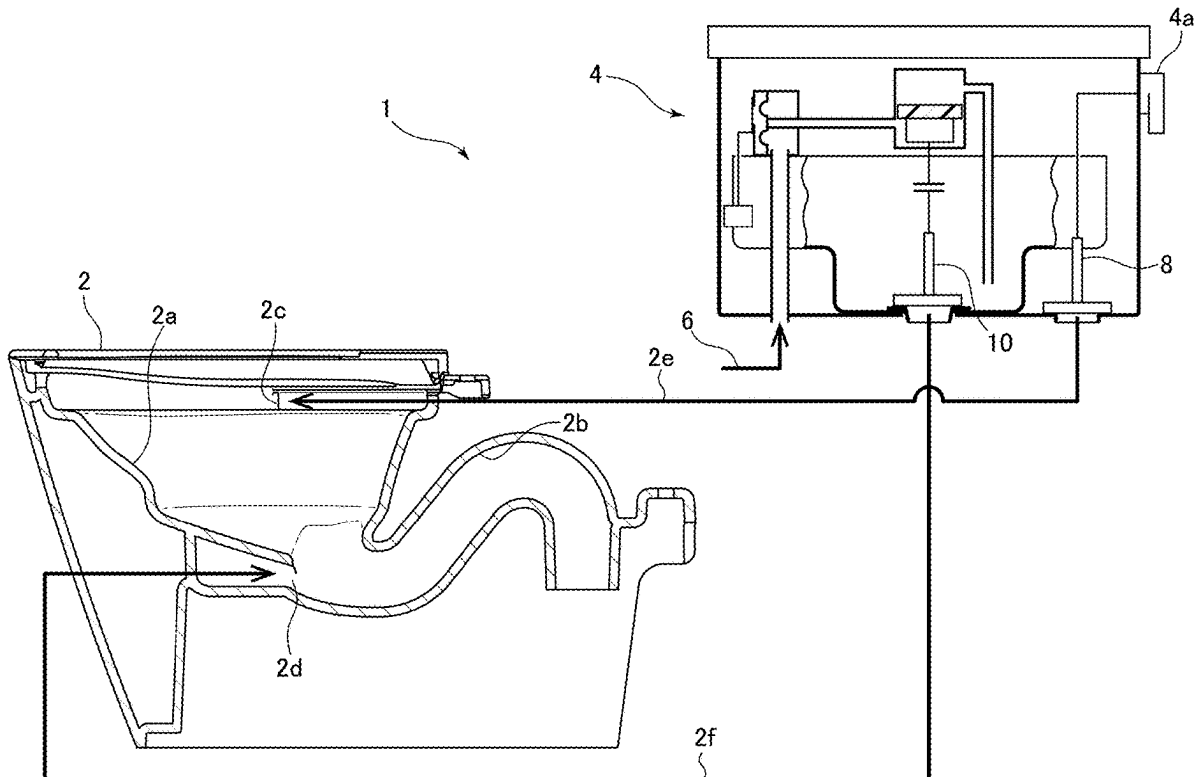
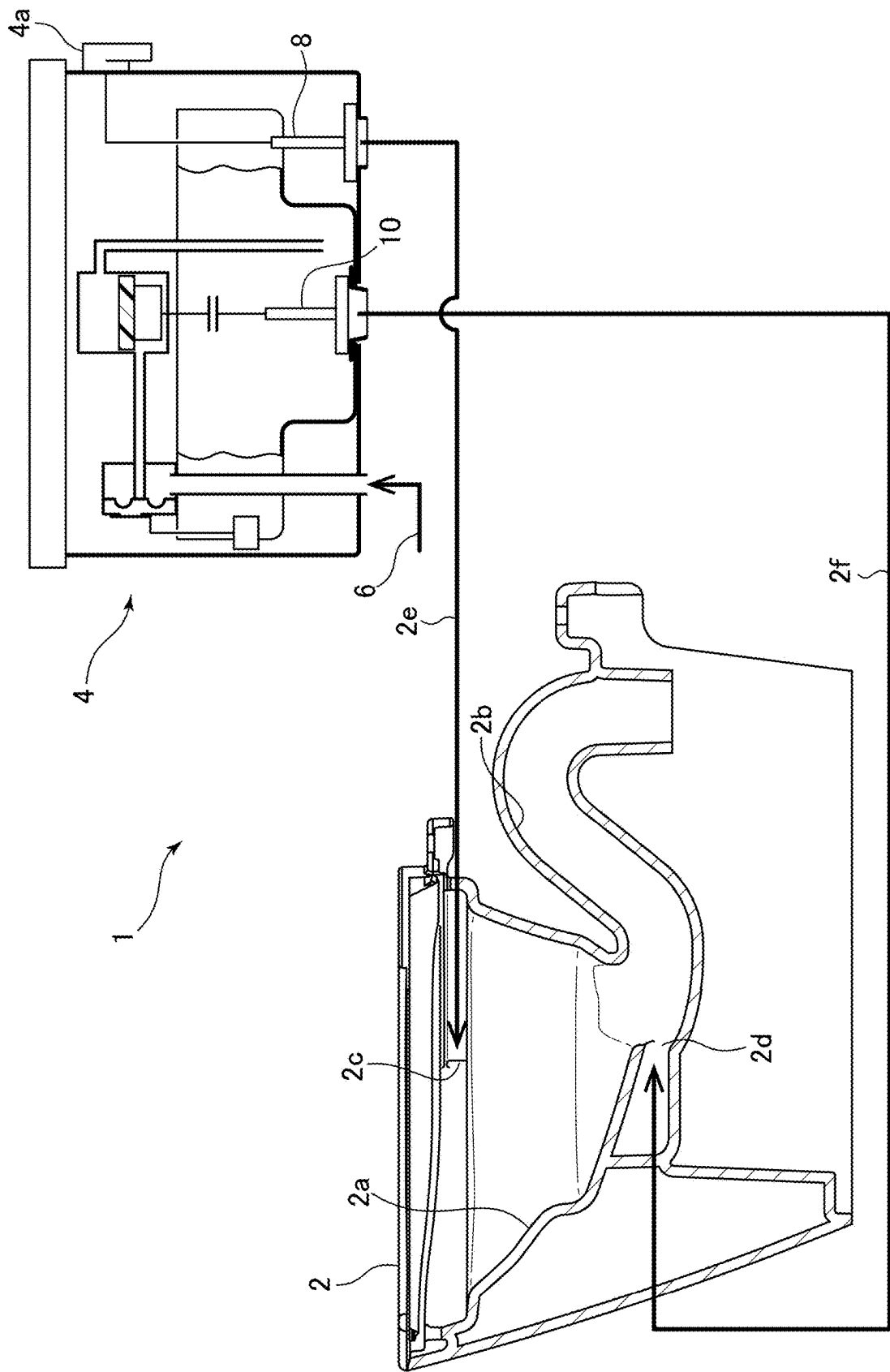


FIG.1



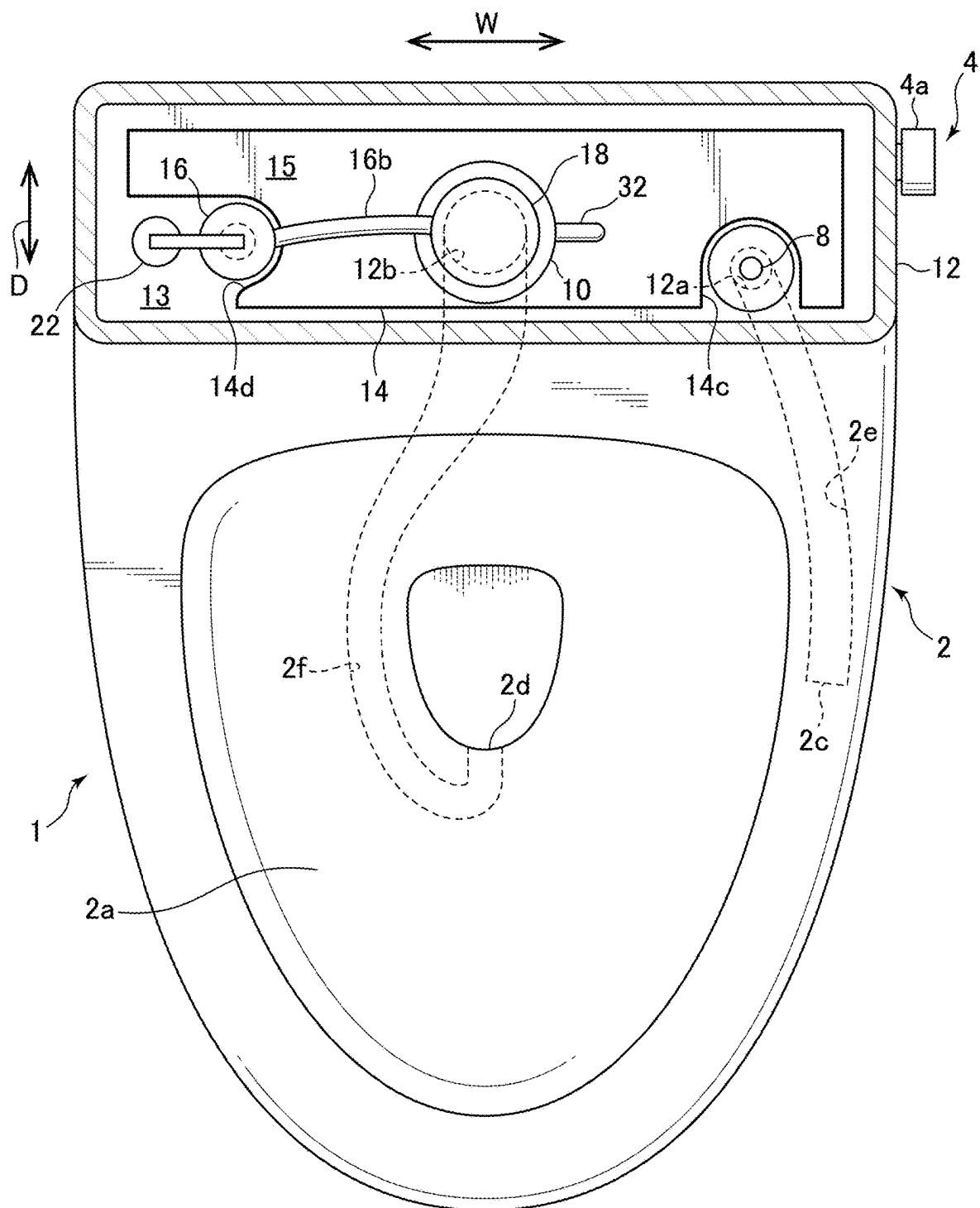


FIG.3

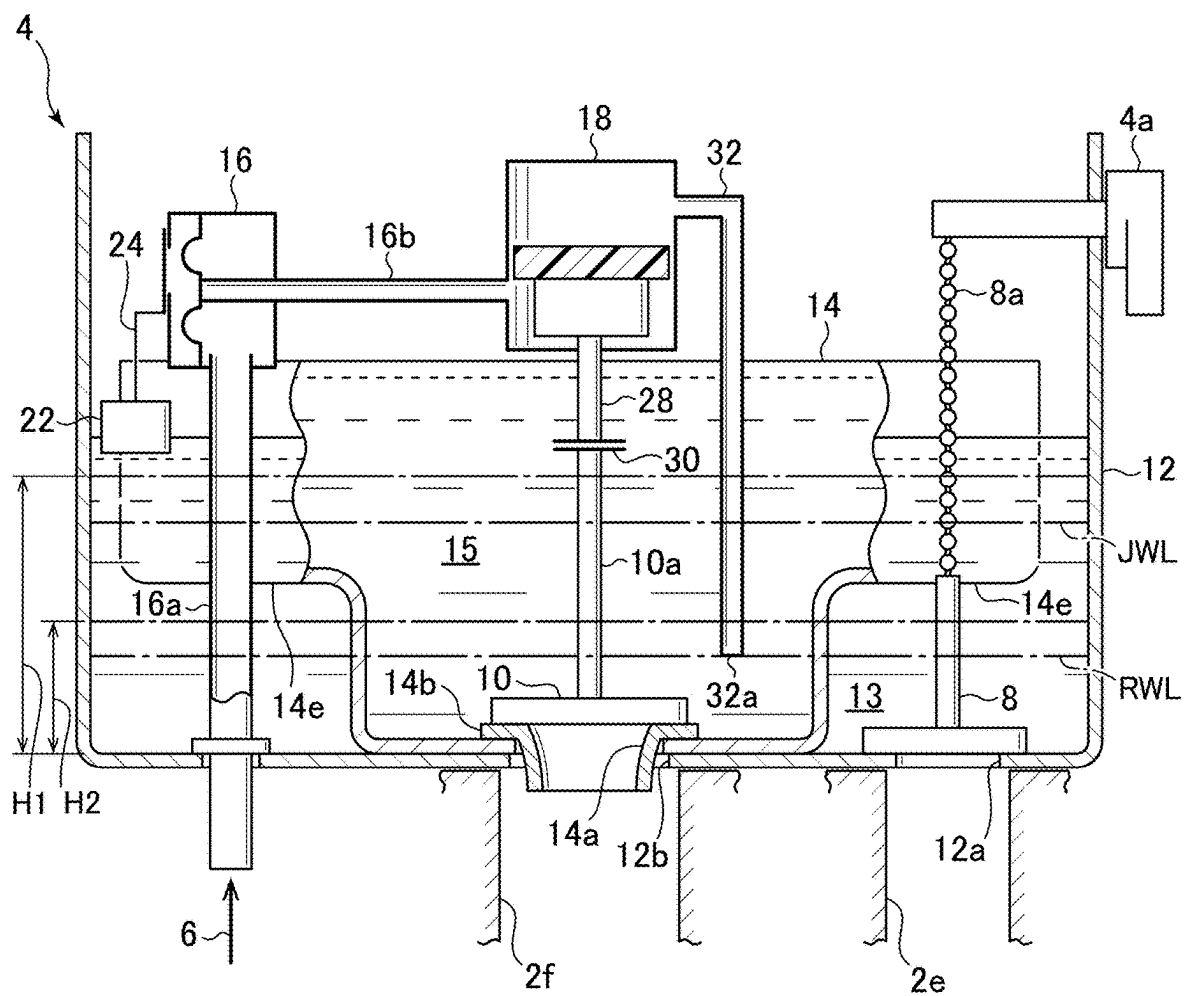


FIG.4

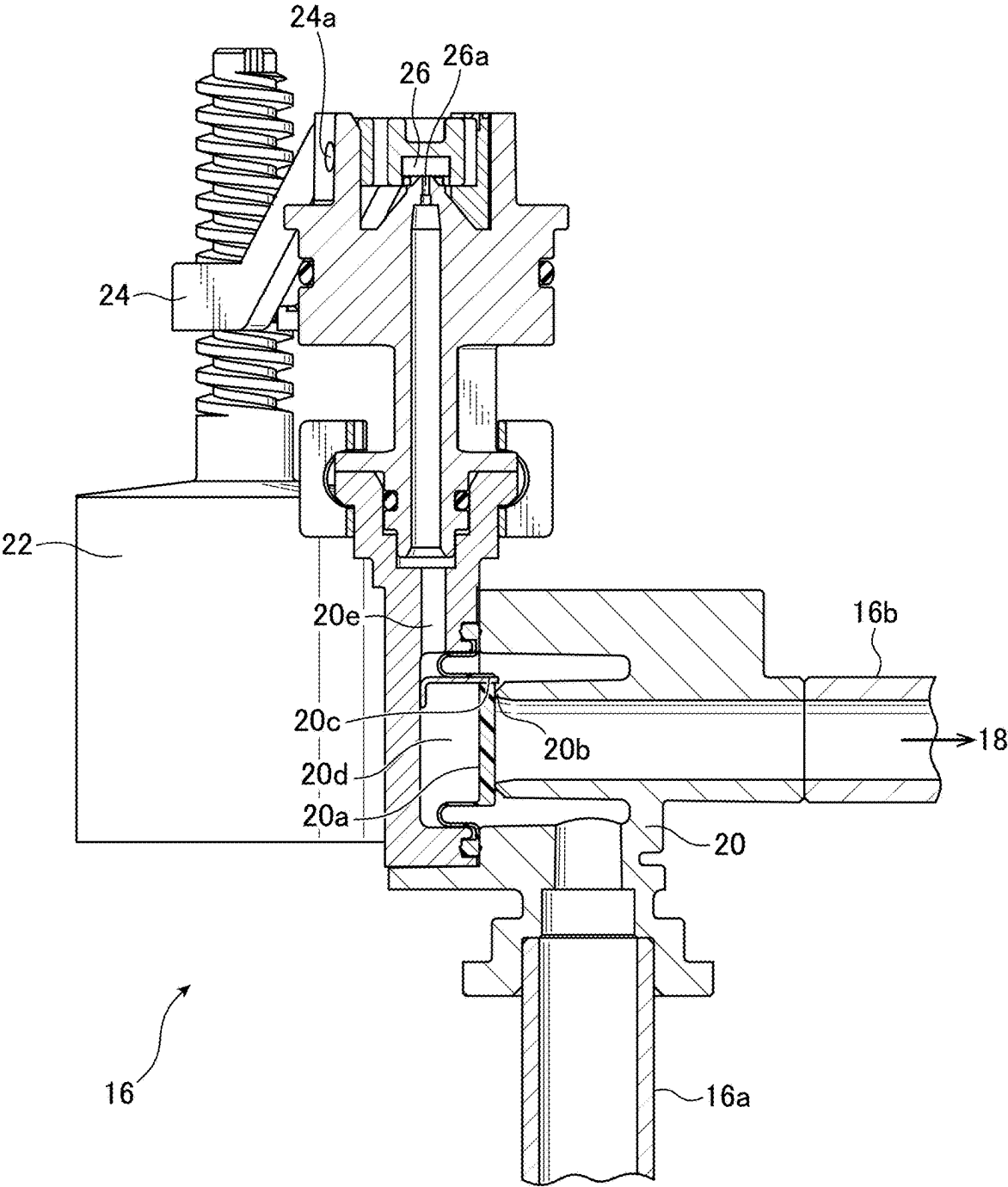


FIG.5

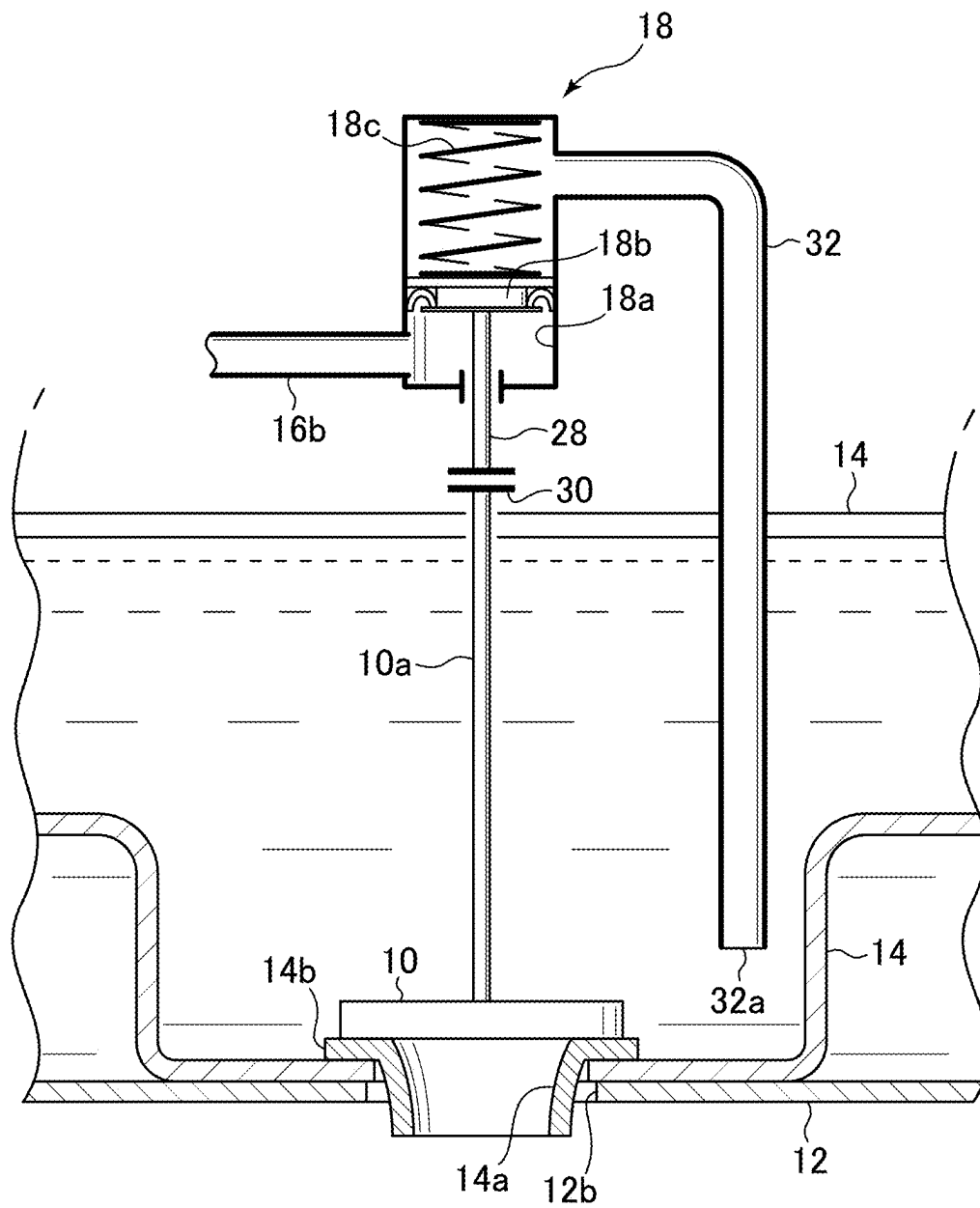


FIG. 6

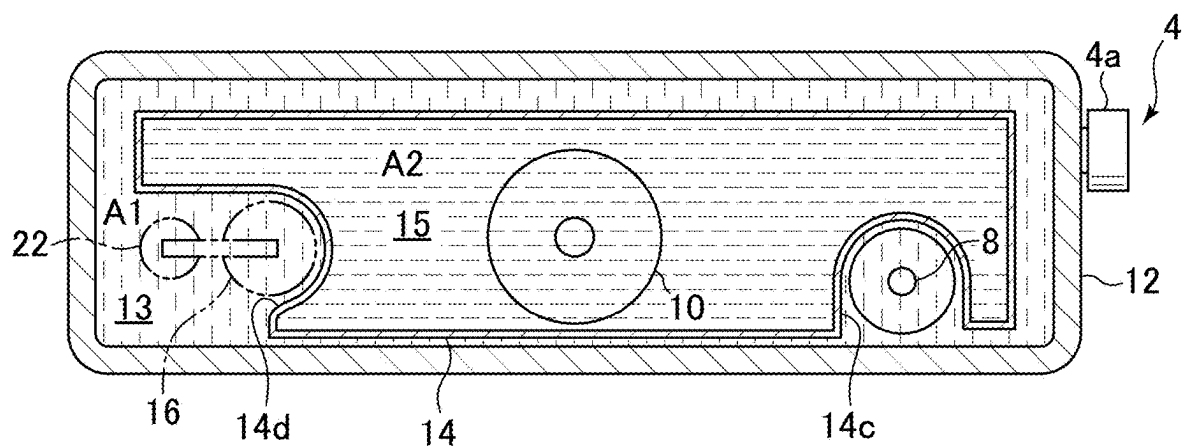


FIG. 7

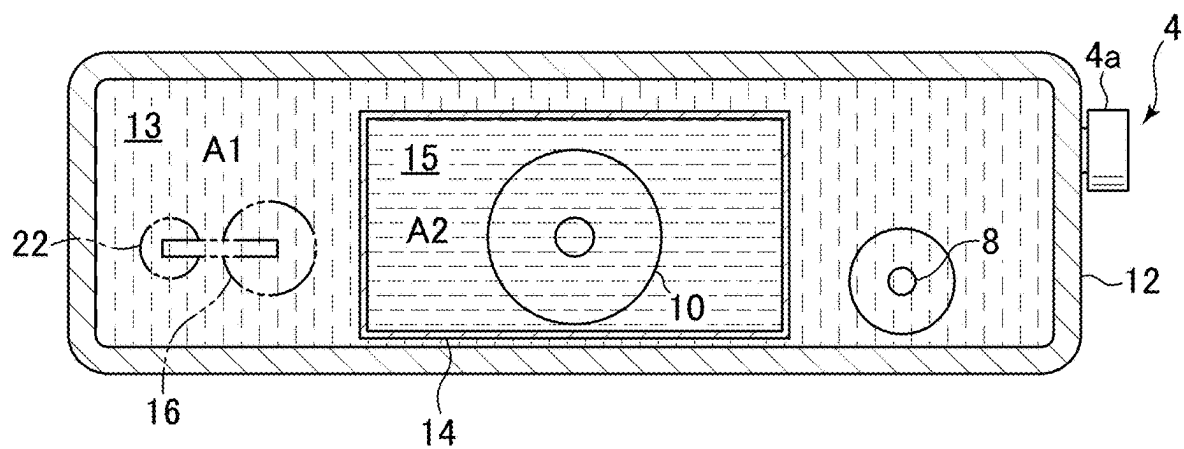
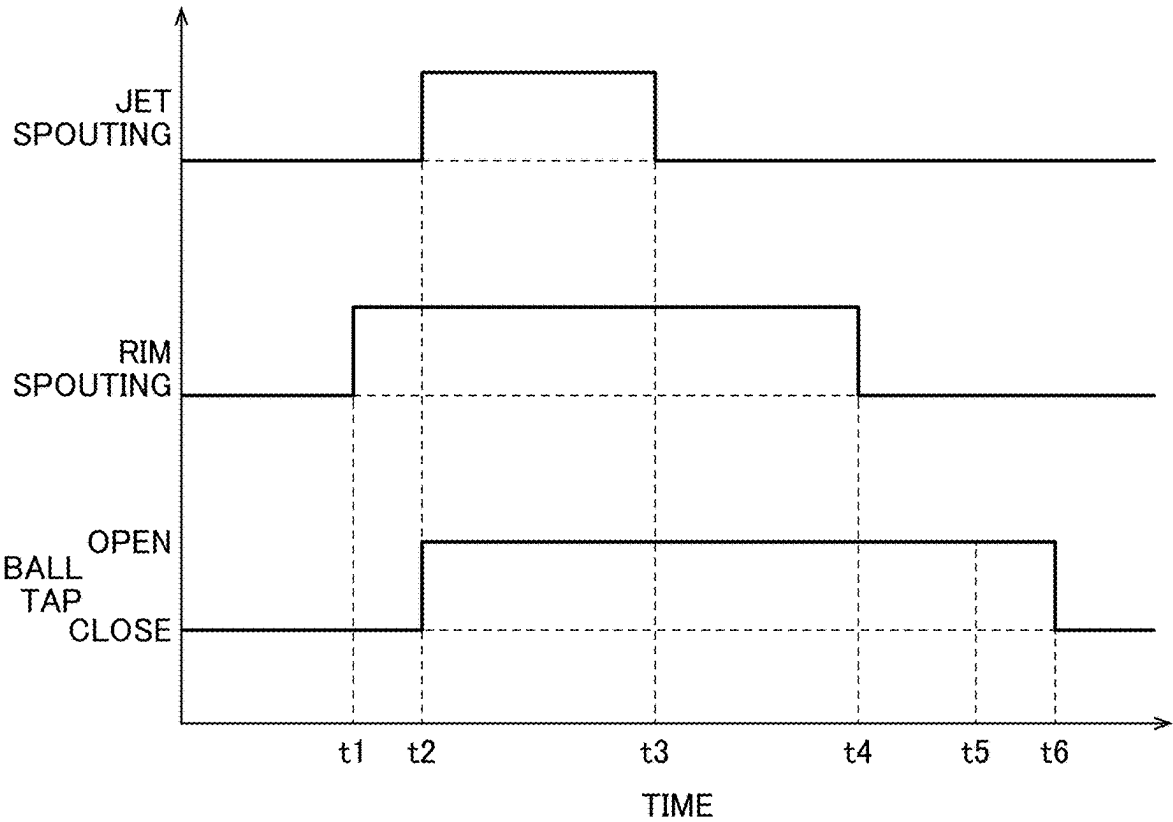


FIG.8



FLUSH WATER TANK DEVICE, AND FLUSH TOILET APPARATUS INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of Japanese Patent Application No. 2024-029551 filed on Feb. 29, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a flush water tank device, particularly to a flush water tank device attached to a flush toilet main body for use to supply flush water, and a flush toilet apparatus including the flush water tank device.

Description of the Related Art

[0003] In Chinese Patent Laid-Open No. CN103967088 (Patent Literature 1), a toilet water tank device is described. In this toilet water tank device, a partition formed by a wall surface vertical to the interior of a water tank is provided, and this partition constitutes a first water tank and a second water tank. The first water tank and the second water tank include a first discharge valve and a second discharge valve, respectively, and are configured so that flush water in the first water tank and flush water in the second water tank are supplied to a flush toilet.

[0004] In the toilet water tank device according to Patent Literature 1, when the first discharge valve is opened, flush water is spouted by a hydraulic head pressure of flush water in the first water tank, and when the second discharge valve is opened, flush water is discharged by a hydraulic head pressure of flush water in the second water tank. However, in the toilet water tank device described in Patent Literature 1, the water tank is simply partitioned to the first water tank and the second water tank, and hence there is a problem that a volume of each water tank and the hydraulic head pressure of flush water in the water tank cannot be sufficiently utilized for toilet flushing.

[0005] Accordingly, an object of the present invention is to provide a flush water tank device capable of sufficiently utilizing a volume in a flush water tank and a hydraulic head pressure of flush water in the flush water tank for toilet flushing, and a flush toilet apparatus including the flush water tank device.

SUMMARY OF THE INVENTION

[0006] To achieve the above object, the present invention provides a flush water tank device which is attached to a flush toilet main body for use to supply flush water, including a flush water tank main body including a first tank part and a second tank part for storing flush water supplied to the flush toilet main body, a rim spouting switch mechanism that switches discharge and stop of flush water in the first tank part so that the flush water in the first tank part is spouted from a rim spout port of the flush toilet main body, and a jet spouting discharge valve that switches discharge and stop of flush water in the second tank part so that the flush water in the second tank part is spouted from a jet spout port of the flush toilet main body, wherein the first tank part and the second tank part are configured to have different cross-

sectional areas depending on a height, and a ratio of the cross-sectional area of the first tank part to the cross-sectional area of the second tank part at a first height is different from a ratio of the cross-sectional area at a second height that is lower than the first height.

[0007] Thus, in the present invention, the first tank part and the second tank part are configured to have different cross-sectional areas depending on the height, and the ratio of the cross-sectional area of the first tank part to the cross-sectional area of the second tank part at the first height is different from the ratio of the cross-sectional area at the second height that is lower than the first height. According to the present invention including this configuration, for example, when the second tank part is configured so that the cross-sectional area of the second tank part is large at the first height and small at the second height, and in the initial stage of flushing, a large amount of flush water can be spouted from the jet spout port at a high hydraulic head pressure. That is, according to the present invention, it is possible to distribute the flush water in the flush water tank main body to the rim spout port and the jet spout port of the flush toilet main body in accordance with a required amount of flush water and water force. Consequently, the volume in the flush water tank main body and the hydraulic head pressure of flush water in the flush water tank main body can be fully utilized for toilet flushing.

[0008] In the present invention, preferably, the second tank part is configured to have a cross-sectional area that is equal to or more than the cross-sectional area of the first tank part at the first height.

[0009] According to the present invention including this configuration, since the second tank part is configured to have a cross-sectional area that is equal to or more than the cross-sectional area of the first tank part at the first height, in the initial stage of flushing in which a water level in the second tank part is high, an instantaneous flow rate of flush water spouted from the jet spout port of the flush toilet main body can be increased, and flushing effect can be enhanced.

[0010] In the present invention, preferably, the second tank part includes a stepped portion provided at a height between the first height and the second height, and the second tank part is configured to have a reduced cross-sectional area below the stepped portion, while the first tank part is configured to have an enlarged cross-sectional area below the stepped portion.

[0011] According to the present invention including this configuration, the second tank part is configured to have the reduced cross-sectional area below the stepped portion, while the first tank part is configured to have the enlarged cross-sectional area below the stepped portion. As a result, in the initial stage of flushing, the instantaneous flow rate from the jet spout port can be increased, and concurrently, a sufficient volume can be acquired in the first tank part at the height that is equal to or less than the height of the stepped portion, so that a sufficient amount of flush water can be allocated to the rim spout port.

[0012] In the present invention, preferably, an inner tank is disposed in the flush water tank main body, the second tank part is formed inside the inner tank, and the first tank part is formed inside the flush water tank main body and outside the inner tank.

[0013] According to the present invention including this configuration, the inner tank is disposed in the flush water tank main body, the second tank part is formed inside the

inner tank, and the first tank part is formed inside the flush water tank main body and outside the inner tank. Therefore, the cross-sectional area of each of the first and second tank parts at each height can be freely set, and the amount of flush water and the hydraulic head pressure can be freely set as required.

[0014] In the present invention, preferably, the jet spouting discharge valve is configured to discharge the flush water in the second tank part so that a water level in the second tank part lowers from a predetermined initial water level to a predetermined jet dead water level, and the stepped portion is provided at a position that is lower than the jet dead water level.

[0015] According to the present invention including this configuration, since the height of the stepped portion in which the cross-sectional area of the second tank part is reduced is set to be lower than the jet dead water level, the flow rate of flush water spouted from the jet spout port can be maintained to be sufficiently high until the spouting is ended.

[0016] In the present invention, preferably, the rim spouting switch mechanism is configured to discharge the flush water in the first tank part so that the water level in the first tank part lowers from the predetermined initial water level to a predetermined rim dead water level, and the stepped portion is provided at a position that is higher than the rim dead water level.

[0017] According to the present invention including this configuration, since the stepped portion in which the cross-sectional area of the first tank part enlarges is provided at a position higher than the rim dead water level, the flush water stored below the stepped portion of the first tank part can be spouted from the rim spout port, and a sufficient amount of flush water spouted from the rim spout port can be acquired.

[0018] In the present invention, preferably, the first tank part and the second tank part are configured so that the ratio of the cross-sectional area is reversed at the first height and the second height.

[0019] According to the present invention including this configuration, since the first tank part and the second tank part are configured so that the ratio of the cross-sectional area is reversed at the first height and the second height, the instantaneous flow rate of flush water spouted from either one of the jet spout port and the rim spout port can be increased. Concurrently, enough flush water spouted from the other port can be acquired.

[0020] Furthermore, the present invention provides a flush toilet apparatus that is flushed with flush water stored in a flush water tank, including a flush toilet main body including a bowl, a rim spout port, and a jet spout port, and the flush water tank device of the present invention that supplies flush water to the rim spout port and the jet spout port of the flush toilet main body.

[0021] According to a flush water tank device of the present invention, and a flush toilet apparatus including the flush water tank device, a volume in a flush water tank and a hydraulic head pressure of flush water in the flush water tank can be sufficiently utilized for toilet flushing.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a block diagram showing an entire configuration of a flush toilet apparatus according to an embodiment of the present invention;

[0023] FIG. 2 is a top view showing a schematic configuration of the flush toilet apparatus according to the embodiment of the present invention;

[0024] FIG. 3 is a front cross-sectional view showing a schematic configuration of a flush water tank device provided in the flush toilet apparatus according to the embodiment of the present invention;

[0025] FIG. 4 is a cross-sectional view showing a structure of a ball tap built in the flush water tank device according to the embodiment of the present invention;

[0026] FIG. 5 is a cross-sectional view showing a structure of a hydraulic drive mechanism built in the flush water tank device according to the embodiment of the present invention;

[0027] FIG. 6 is a horizontal cross-sectional view of the flush water tank device according to the embodiment of the present invention at a height H1 of FIG. 3;

[0028] FIG. 7 is a horizontal cross-sectional view of the flush water tank device according to the embodiment of the present invention at a height H2 of FIG. 3; and

[0029] FIG. 8 is a time chart showing the operation of the flush toilet apparatus according to the embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0030] Next, with reference to the accompanying drawings, a flush water tank device according to an embodiment of the present invention and a flush toilet apparatus including the same will be described.

[0031] FIG. 1 is a block diagram showing an entire configuration of the flush toilet apparatus according to the embodiment of the present invention. FIG. 2 is a top view showing a schematic configuration of the flush toilet apparatus according to the embodiment of the present invention. FIG. 3 is a front cross-sectional view showing a schematic configuration of a flush water tank device provided in the flush toilet apparatus according to the embodiment of the present invention. FIG. 4 is a cross-sectional view showing a structure of a ball tap built in the flush water tank device according to the embodiment of the present invention. FIG. 5 is a cross-sectional view showing a structure of a hydraulic drive mechanism built in the flush water tank device according to the embodiment of the present invention. FIG. 6 is a horizontal cross-sectional view of the flush water tank device at a height H1 of FIG. 3. FIG. 7 is a horizontal cross-sectional view of the flush water tank device at a height H2 of FIG. 3.

[0032] As shown in FIGS. 1 and 2, a flush toilet apparatus 1 according to the embodiment of the present invention includes a flush toilet main body 2 and a flush water tank device 4 disposed at the rear of the flush toilet main body 2. The flush toilet apparatus 1 of the present embodiment is configured to be flushed by operating a lever handle 4a provided in the flush water tank device 4 after use.

[0033] The flush toilet main body 2 includes a bowl 2a and a discharge trap conduit 2b extending from a lower part of the bowl 2a. Furthermore, a rim spout port 2c is provided in a top edge portion of the bowl 2a, and a jet spout port 2d is provided in the lower part of the bowl 2a. During toilet flushing, flush water is spouted from each of the rim spout port 2c and the jet spout port 2d at a predetermined timing, and a waste receiving surface of the bowl 2a is flushed, while waste and flush water in the bowl 2a are drained to the

discharge trap conduit 2b. The waste and flush water drained to the discharge trap conduit 2b are drained through a discharge socket (not shown) to a sewage pipe (not shown).

[0034] Flush water is supplied to the flush water tank device 4 from a water supply source 6 such as a water supply, and the supplied flush water is stored in the flush water tank device 4. Furthermore, in the flush water tank device 4, a rim spouting discharge valve 8 that is a rim spouting switch mechanism, and a jet spouting discharge valve 10 are built and are configured to open and close discharge ports, respectively, provided in a bottom portion of the flush water tank device 4.

[0035] In the present embodiment, by opening the rim spouting discharge valve 8, flush water is spouted from the rim spout port 2c through a rim conduit 2e formed inside the flush toilet main body 2. Furthermore, by opening the jet spouting discharge valve 10, flush water is spouted from the jet spout port 2d through a jet conduit 2f formed inside the flush toilet main body 2.

[0036] Next, with reference to FIGS. 2 and 3, an internal structure of the flush water tank device 4 will be described.

[0037] As shown in FIGS. 2 and 3, the flush water tank device 4 includes a flush water tank main body 12, an inner tank 14 disposed inside the flush water tank main body, the rim spouting discharge valve 8 disposed in the flush water tank main body 12, the jet spouting discharge valve 10 disposed in the inner tank 14, a ball tap 16 that is a water supply valve, and a hydraulic drive mechanism 18.

[0038] The flush water tank main body 12 and the inner tank 14 are containers configured to store flush water to be supplied to the flush toilet main body 2. In the present embodiment, the flush water tank main body 12 is made of ceramic, and the inner tank 14 disposed inside the flush water tank main body 12 is made of resin. In the present embodiment, a space inside the flush water tank main body 12 and outside the inner tank 14 constitutes a first tank part 13, and a space in the inner tank 14 constitutes a second tank part 15. Therefore, in the present embodiment, the first tank part 13 and the second tank part 15 for storing flush water supplied to the flush toilet main body 2 are provided inside the flush water tank main body 12.

[0039] Furthermore, as shown in FIG. 3, a first drain port 12a and a second drain port 12b are provided on a bottom surface of the flush water tank main body 12, and in the present embodiment, these drain ports are formed in a circular shape. Here, the flush water stored in the first tank part 13 (inside the flush water tank main body 12 and outside the inner tank 14) flows through the first drain port 12a into the rim conduit 2e of the flush toilet main body 2 and is spouted from the rim spout port 2c. Furthermore, the flush water stored in the second tank part 15 (in the inner tank 14) flows through the second drain port 12b into the jet conduit 2f of the flush toilet main body 2 and is spouted from the jet spout port 2d.

[0040] The inner tank 14 is disposed on the bottom surface of the flush water tank main body 12, and the lower part of the inner tank 14 is submerged in the flush water stored in the flush water tank main body 12. Furthermore, a circular drain port 14a is formed on the bottom surface of the inner tank 14. The drain port 14a of the inner tank 14 is disposed concentrically to match the second drain port 12b provided in the flush water tank main body 12. That is, in the present embodiment, the center of the circular second drain port 12b and the circular drain port 14a coincide in top view. There-

fore, the flush water in the inner tank 14 flows into the jet conduit 2f of the flush toilet main body 2 through the drain port 14a of the inner tank 14 and the second drain port 12b of the flush water tank main body 12.

[0041] Furthermore, as shown in FIG. 3, in the present embodiment, the drain port 14a of the inner tank 14 is composed of a drain port forming member 14b composed separately from a main body part of the inner tank 14. The drain port forming member 14b is a tubular member and is water-tightly attached to the bottom surface of the inner tank 14 to form the drain port 14a inside. Furthermore, a seat surface is provided at an upper end of the drain port forming member 14b, and the drain port 14a is closed when the jet spouting discharge valve 10 is seated on the seat surface. Therefore, in the present embodiment, the seat surface on which the jet spouting discharge valve 10 is seated is composed of a member separate from the inner tank 14.

[0042] Furthermore, as shown in FIG. 2, the flush water tank main body 12 of the flush water tank device 4 is formed in a substantially rectangular shape that is long in a width direction of the flush toilet main body 2 (arrow W direction in FIG. 2) and short in a depth direction (arrow D direction in FIG. 2) in top view. Furthermore, the inner tank 14 disposed in the flush water tank main body 12 is also formed to be long in the width direction and short in the depth direction in top view and is irregularly shaped to avoid the rim spouting discharge valve 8 and the ball tap 16.

[0043] Specifically, in the present embodiment, the ball tap 16 is disposed in a front left end portion (lower left corner in FIG. 2) in the flush water tank main body 12, and the rim spouting discharge valve 8 is disposed in a front right end portion (lower right corner in FIG. 2). The inner tank 14 is provided with cutouts 14c and 14d in opposite front end portions in the width direction in top view to avoid the ball tap 16 and the rim spouting discharge valve 8. That is, in the flush water tank main body 12, the cutout 14c that is U-shaped is provided in the inner tank 14 in a portion in which the rim spouting discharge valve 8 is disposed, and the cutout 14d is provided in a portion in which the ball tap 16 is disposed, so that the valve and ball tap do not interfere with the inner tank 14.

[0044] On the other hand, as shown in FIG. 3, a stepped portion 14e is formed on each side of the inner tank 14 in the width direction, and the inner tank is configured to have a reduced width below the stepped portions 14e. That is, the inner tank 14 is substantially formed in a T-shape in front view so that a horizontal cross-sectional area A2 (FIGS. 6 and 7) is constant above and below the stepped portion 14e. In other words, the second tank part 15, which is an inner space of the inner tank 14, and the first tank part 13, which is an outer space, are configured to have different horizontal cross-sectional areas depending on the height. That is, the second tank part 15 is configured to have a reduced horizontal cross-sectional area A2 below the stepped portion 14e, while the first tank part 13 is configured to have an enlarged horizontal cross-sectional area A1 below the stepped portion 14e.

[0045] As a result, in the present embodiment, a ratio of the horizontal cross-sectional area of the first tank part 13 to that of the second tank part 15 at a first height H1 above the stepped portion 14e ($A1/A2$ in FIG. 6) and a ratio of the horizontal cross-sectional area of the first tank part 13 to that of the second tank part 15 at a second height H2 below the first height H1 and the stepped portion 14e ($A1/A2$ in FIG.

7) are different. In the present embodiment, as shown in FIG. 6, in the first height H1 above the stepped portion 14e, the horizontal cross-sectional area A2 of the second tank part 15 is larger than the horizontal cross-sectional area A1 of the first tank part 13. As shown in FIG. 7, in the second height H2 below the stepped portion 14e, the horizontal cross-sectional area A1 of the first tank part 13 is larger than the horizontal cross-sectional area A2 of the second tank part 15. Therefore, in the present embodiment, the ratio of the horizontal cross-sectional area of the first tank part 13 to that of the second tank part 15 is reversed at the first height H1 and the second height H2.

[0046] Furthermore, during actual use of the flush water tank device 4, flush water is stored to a level above the stepped portion 14e of the inner tank 14, in the second tank part 15. Therefore, in the present embodiment, the inner tank 14 is submerged in the flush water stored in the first tank part 13 up to the level above the stepped portion 14e. Then, in the present embodiment, by configuring a large upper part of the second tank part 15, a large amount of flush water having large potential energy is stored in the second tank part 15. Consequently, the hydraulic head pressure of the flush water drained from the second tank part 15 can be increased, and the instantaneous flow rate of the flush water spouted from the jet spout port 2d can be increased.

[0047] Furthermore, by irregularly shaping the inner tank 14 with resin, sufficient volume is acquired in the inner tank 14 while suppressing a dimension of the flush water tank device 4 in the depth direction. Additionally, the inner tank 14 made of resin can be configured with a thinner wall surface than an inner tank made of ceramic, and hence the volume in the flush water tank main body 12 can be effectively utilized.

[0048] Next, the rim spouting discharge valve 8, which is the rim spouting switch mechanism, is a valve body disposed to open and close the first drain port 12a provided in the flush water tank main body 12, and the first drain port 12a is opened by pulling the rim spouting discharge valve 8 upward. Consequently, the flush water in the first tank part 13 is drained to the rim conduit 2e (FIG. 1) of the flush toilet main body 2 and spouted from the rim spout port 2c. Therefore, the rim spouting discharge valve 8, which is the rim spouting switch mechanism, switches the discharge and stop of the flush water stored in the first tank part 13 to the flush toilet main body 2.

[0049] In the present embodiment, a bead chain 8a (FIG. 3) coupled to the rim spouting discharge valve 8 is pulled by a user rotating and operating the lever handle 4a provided in the flush water tank device 4, and the rim spouting discharge valve 8 is pulled upward. As a modification, the present invention may include a configuration in which the rim spouting discharge valve 8 is pulled up based on a control signal from a remote controller (not shown) or a detection signal from a human sensor (not shown), to perform flushing.

[0050] Furthermore, in the present embodiment, the rim spouting discharge valve 8 is provided as the rim spouting switch mechanism, and a configuration other than the rim spouting discharge valve 8 may be used as the rim spouting switch mechanism. For example, a jet pump (not shown) that causes flush water stored in the first tank part 13 to flow into the flush toilet main body may be used as the rim spouting switch mechanism. In this case, a jet pump nozzle (not shown) is disposed in the first tank part 13, and through

this nozzle, part of the flush water supplied via a water supply valve such as the ball tap 16 is injected. By injecting flush water from the nozzle in the first tank part 13, the flush water stored in the first tank part 13 can be entrained, and the flush water in the first tank part 13 can flow into the flush toilet main body by jet pump action. In addition to this jet pump (not shown), an arbitrary configuration that switches discharge and stop of flush water to the flush toilet main body 2 may be used as the rim spouting switch mechanism.

[0051] Next, the jet spouting discharge valve 10 is a valve body disposed to open and close the drain port 14a provided in the inner tank 14, and by pulling the jet spouting discharge valve 10 upward, the jet spouting discharge valve 10 is unseated from the seat surface of the drain port 14a, to open the drain port 14a. Consequently, the flush water in the second tank part 15 is drained from the drain port 14a, flows through the second drain port 12b into the jet conduit 2f (FIG. 1) of the flush toilet main body 2, and is spouted from the jet spout port 2d. Therefore, the jet spouting discharge valve 10 switches the discharge and stop of the flush water stored in the second tank part 15 to the flush toilet main body 2.

[0052] In the present embodiment, the jet spouting discharge valve 10 is configured to be pulled up from the drain port 14a by the hydraulic drive mechanism 18. That is, the jet spouting discharge valve 10 is a valve body including a valve shaft 10a extending upward, and the valve shaft 10a is pulled up by the hydraulic drive mechanism 18. Then, when pulled up to a predetermined height, the jet spouting discharge valve 10 is disconnected from the hydraulic drive mechanism 18, and gently descends to close the drain port 14a. The configuration of the hydraulic drive mechanism 18 will be described later, but the present invention may be applied to a flush water tank device that is not provided with the hydraulic drive mechanism 18.

[0053] Furthermore, the ball tap 16, which is a water supply valve, includes an inflow pipe 16a through which flush water supplied from the water supply source 6 flows into the ball tap, and is configured to switch supply and stop of flush water to be stored in the first tank part 13 and the second tank part 15.

[0054] Next, with reference to FIG. 4, the configuration of the ball tap 16 will be described.

[0055] As shown in FIG. 4, the ball tap 16 includes a main body 20 to which the inflow pipe 16a and an outflow pipe 16b are connected, a main valve body 20a disposed in the main body 20, a valve seat 20b on which the main valve body 20a is seated, an arm 24 rotated by a float 22, and a pilot valve 26 moved by the rotation of the arm 24. That is, the ball tap 16 includes the float 22 that operates in conjunction with a water level in the flush water tank device 4, and the float 22 is configured to supply flush water to the hydraulic drive mechanism 18 when lowering to a predetermined position.

[0056] The main body 20 is a member having a lower part including a connecting portion to the inflow pipe 16a and having one side including a connecting portion to the outflow pipe 16b. Furthermore, the valve seat 20b is formed inside the main body 20, and the valve seat 20b communicates with the outflow pipe 16b connected to the connecting portion. Furthermore, inside the main body 20, the main valve body 20a is disposed to open and close the valve seat 20b and is configured so that when the valve is opened, tap water flowing from the inflow pipe 16a flows through the

valve seat **20b**, to flow out to the outflow pipe **16b**. The outflow pipe **16b** is then connected to the hydraulic drive mechanism **18**.

[0057] The main valve body **20a** is a diaphragm valve body that is generally disc-shaped and is attached to the main body **20** so that the main valve body can be seated on and unseated from the valve seat **20b**. Furthermore, a bleed hole **20c** is provided in a peripheral edge portion of the main valve body **20a**. Furthermore, in the main valve body **20a**, a pressure chamber **20d** is formed on a side of the main valve body **20a** opposite to the valve seat **20b** (left side in FIG. 4). That is, the pressure chamber **20d** is defined by an inner wall surface of the main body **20** and the main valve body **20a**, and when the pressure in the pressure chamber **20d** increases, the main valve body **20a** is pressed against the valve seat **20b** and seated on the valve seat **20b** by this pressure.

[0058] Furthermore, a pressure passage **20e** extends upward to communicate with the pressure chamber **20d** provided in the main body **20**, and a pilot valve port **26a** is provided at an upper end of the pressure passage **20e**. The pilot valve port **26a** is opened upward and is configured to be opened and closed by the pilot valve **26**.

[0059] On one hand, the float **22** is supported by the arm **24**, and the arm **24** is rotatably supported by a support shaft **24a**. Furthermore, the pilot valve **26** is coupled to the arm **24**, and the pilot valve **26** is configured to move in an up-down direction together with the rotation of the arm **24**. In the present embodiment, the float **22** is disposed in the second tank part **15** and moved vertically depending on the water level in the second tank part **15**. Consequently, when the water level in the second tank part **15** is raised to or above a predetermined water level, the float **22** is pushed upward, and the pilot valve **26** is accordingly moved downward and seated on the pilot valve port **26a**, to close the pilot valve port. On the other hand, when the flush water in the second tank part **15** is discharged to lower the water level, the float **22** moves downward, and the pilot valve **26** moves upward, to open the pilot valve port **26a**. Consequently, when toilet flushing is on standby in a state in which the water level in the flush water tank main body **12** is higher than the predetermined water level, the pilot valve port **26a** of the main body **20** is closed.

[0060] Furthermore, tap water flowing through the inflow pipe **16a** into the main body **20** flows into an annular space around the valve seat **20b** and flows from here through the bleed hole **20c** of the main valve body **20a** into the pressure chamber **20d**. In this state in which the pilot valve port **26a** is closed by the pilot valve **26**, there is no path for tap water flowing from the bleed hole **20c** into the pressure chamber **20d** to flow out, and the pressure in the pressure chamber **20d** rises. When the pressure in the pressure chamber **20d** rises in this way, the main valve body **20a** is pressed toward the valve seat **20b** (rightward in FIG. 4) by this pressure, and the valve seat **20b** is closed by the main valve body **20a**.

[0061] In contrast, when the rim spouting discharge valve **8** is opened by the flushing operation and the water level in the first tank part **13** becomes lower than the predetermined water level, the float **22** descends, and the pilot valve **26** moves upward, to open the pilot valve port **26a**. When the pilot valve port **26a** is opened, water in the pressure chamber **20d** flows out of the pilot valve port **26a** and the pressure in the pressure chamber **20d** decreases. Consequently, the main valve body **20a** is moved to be pulled away from the valve

seat **20b** (leftward in FIG. 4), to open the valve seat **20b**. Consequently, in this state in which the pilot valve port **26a** is opened, the pressure in the pressure chamber **20d** does not increase, so that the valve seat **20b** is opened.

[0062] Next, with reference to FIG. 5, the configuration of the hydraulic drive mechanism **18** will be described.

[0063] The hydraulic drive mechanism **18** is configured to drive the jet spouting discharge valve **10** using a water supply pressure of the flush water supplied from the water supply to the flush water tank device **4**. Specifically, the hydraulic drive mechanism **18** includes a cylinder **18a** into which water supplied from the ball tap **16** flows, a piston **18b** slidably disposed in the cylinder **18a**, and a rod **28** that protrudes from a lower end of the cylinder **18a** to drive the jet spouting discharge valve **10**. Furthermore, a spring **18c** is disposed inside the cylinder **18a**, to bias the piston **18b** downward, and a packing is attached to the piston **18b**, to acquire watertightness between an inner wall surface of the cylinder **18a** and the piston **18b**. In addition, a clutch mechanism **30** is provided at a lower end of the rod **28**, and the rod **28** and the valve shaft **10a** of the jet spouting discharge valve **10** are coupled/decoupled by the clutch mechanism **30**.

[0064] The cylinder **18a** is a cylindrical member, has an axis oriented in a vertical direction and can slidably receive the piston **18b** inside. Furthermore, the outflow pipe **16b** extending from the ball tap **16** is connected to a lower end of the cylinder **18a**, and the flush water flowing out of the ball tap **16** flows into the cylinder **18a**. Consequently, the piston **18b** in the cylinder **18a** is pushed up against a biasing force of the spring **18c** by the water flowing into the cylinder **18a**.

[0065] In addition, an outflow hole is provided in an upper end of the cylinder **18a**, and a water supply pipe **32** is connected to the outflow hole. Therefore, when water flows into the cylinder **18a** through the outflow pipe **16b** connected to a lower part of the cylinder **18a**, the piston **18b** is pushed upward from the lower part of the cylinder **18a**. Then, when the piston **18b** is pushed up to a position above the outflow hole, the water flowing into the cylinder **18a** flows out of the outflow hole to the water supply pipe **32**.

[0066] The water supply pipe **32** is a pipe extending downward from the outflow hole provided in the upper end of the cylinder **18a**. The flush water flowing out of the cylinder **18a** flows into the second tank part **15** from a water supply port **32a** at a lower end of the water supply pipe **32**. The water supply port **32a** at the lower end of the water supply pipe **32** is located below the water surface of the flush water in the second tank part **15** on standby. Therefore, the ball tap **16** is configured to cause flush water to flow into the second tank part **15** from the water supply port **32a** provided below the water surface of the second tank part **15**.

[0067] The rod **28** is a bar-shaped member connected to a lower surface of the piston **18b** and extends to protrude downward from the cylinder **18a** via a through hole formed on a bottom surface of the cylinder **18a**. Furthermore, the valve shaft **10a** of the jet spouting discharge valve **10** is connected to the lower end of the rod **28** via the clutch mechanism **30**, and the rod **28** couples the piston **18b** and the jet spouting discharge valve **10**. Consequently, when water flows into the cylinder **18a** to push up the piston **18b**, the rod **28** connected to the piston **18b** lifts the jet spouting discharge valve **10** upward, and the jet spouting discharge valve **10** is opened.

[0068] Furthermore, a gap is provided between the rod 28 protruding from the bottom of the cylinder 18a and the inner wall of the through hole of the cylinder 18a, and part of the water flowing into the cylinder 18a flows out of this gap. The water flowing out of the gap flows into the second tank part 15. This gap is comparatively narrow and has a large flow channel resistance. Therefore, even if water flows out of the gap, the pressure in the cylinder 18a increases due to the water flowing through the outflow pipe 16b into the cylinder 18a, and the piston 18b is pushed up against the biasing force of the spring 18c.

[0069] Furthermore, the clutch mechanism 30 removably couples the rod 28 and the jet spouting discharge valve 10. The clutch mechanism 30 is configured to disconnect the valve shaft 10a of the jet spouting discharge valve 10 from the rod 28, when the jet spouting discharge valve 10 is lifted up together with the rod 28 along a predetermined distance. While the clutch mechanism 30 is disconnected, the jet spouting discharge valve 10 is no longer linked to the movement of the piston 18b and the rod 28, and the jet spouting discharge valve 10 descends with decrease in water level in the second tank part 15, to close the drain port 14a of the inner tank 14.

[0070] Next, newly with reference to FIG. 8, an operation of the flush toilet apparatus 1 according to the embodiment of the present invention will be described.

[0071] FIG. 8 is a time chart showing the operation of the flush toilet apparatus 1 according to the embodiment of the present invention, and shows a state of jet spouting, a state of rim spouting, and a state of the ball tap in order from the top.

[0072] First, in a standby state for toilet flushing, the first drain port 12a of the flush water tank main body 12 and the drain port 14a of the inner tank 14 are closed by the rim spouting discharge valve 8 and the jet spouting discharge valve 10, respectively. Furthermore, in the standby state, flush water is stored up to a predetermined initial water level in each of the first tank part 13 and the second tank part 15. In the present embodiment, the initial water level of the first tank part 13 is set to a position higher than the stepped portion 14e of the inner tank 14. Thus, in a state in which the first tank part 13 is at the initial water level, the pilot valve port 26a of the main body 20 (FIG. 4) of the ball tap 16 is closed, and the valve seat 20b is closed by the main valve body 20a. On the other hand, in the present embodiment, the initial water level of the second tank part 15 is almost as high as the upper edge of the inner tank 14 and is at a position higher than the initial water level of the first tank part 13.

[0073] Next, at time t1 in FIG. 8, when the user rotates and operates the lever handle 4a (FIG. 3) of the flush water tank device 4 to perform toilet flushing, the bead chain 8a connected to the handle pulls up the rim spouting discharge valve 8. Consequently, the rim spouting discharge valve 8 is pulled away from the first drain port 12a, and the first drain port 12a is opened. When the first drain port 12a is opened, the flush water stored in the second tank part 15 flows from the first drain port 12a into the rim conduit 2e (FIG. 2) and is spouted from the rim spout port 2c. By rim spouting from the rim spout port 2c, a circulating flow is formed on the waste receiving surface of the bowl 2a, and the waste receiving surface is flushed.

[0074] As the flush water is drained from the first drain port 12a, the water level in the second tank part 15 decreases. Consequently, the float 22 of the ball tap 16

lowers to open the pilot valve 26 (FIG. 4). This decreases the pressure in the pressure chamber 20d, to open the main valve body 20a, and flush water is supplied through the outflow pipe 16b to the hydraulic drive mechanism 18.

[0075] When flush water is supplied to the hydraulic drive mechanism 18, the flush water flowing into the cylinder 18a (FIG. 5) pushes up the piston 18b against the biasing force of the spring 18c. Consequently, the rod 28 coupled to the piston 18b pulls up the valve shaft 10a of the jet spouting discharge valve 10, and the drain port 14a of the inner tank 14 is opened. That is, the jet spouting discharge valve 10 is driven by the water supply pressure of tap water supplied via the ball tap 16 and is opened.

[0076] At time t2 in FIG. 8, the drain port 14a is opened, and the flush water stored in the second tank part 15 flows through the drain port 14a and the second drain port 12b into the jet conduit 2f (FIG. 2) and is spouted from the jet spout port 2d. The jet spouting from the jet spout port 2d fills the discharge trap conduit 2b with water and induces siphon action. Due to occurrence of the siphon action, retained water and waste in the bowl 2a are suctioned into the discharge trap conduit 2b and drained to the sewage pipe (not shown).

[0077] Here, in a state in which the drain port 14a is closed, the interior of the jet conduit 2f is filled with flush water to a level of pooled water surface of the bowl 2a, and no flush water is in a space above this pooled water surface. Therefore, in a state before the jet spouting discharge valve 10 is opened, no flush water is at an upstream end in the jet conduit 2f.

[0078] When the jet spouting discharge valve 10 is opened, the space in which there is no flush water in the jet conduit 2f is filled in a brief time, and the flush water can be spouted from the jet spout port 2d at an early stage. In particular, according to the present embodiment, the cross-sectional area (cross-sectional area A2 in FIG. 6) of the inner tank 14 above the stepped portion 14e is larger than the cross-sectional area (cross-sectional area A2 in FIG. 7) below the stepped portion, and hence the second tank part 15 includes a large volume at a high position. Consequently, a large amount of flush water having a high hydraulic head pressure can flow into the jet conduit 2f, and the instantaneous flow rate of the flush water spouted from the jet spout port 2d can be increased.

[0079] When the jet spouting discharge valve 10 is pulled up to the predetermined height together with the piston 18b of the hydraulic drive mechanism 18, the valve shaft 10a of the jet spouting discharge valve 10 is disconnected from the rod 28 by the clutch mechanism 30 (FIG. 5). Consequently, the jet spouting discharge valve 10 descends toward the drain port 14a as the water level in the second tank part 15 decreases. Then, at time t3 in FIG. 8, when the water level in the second tank part 15 decreases to a predetermined jet dead water level JWL (FIG. 3), the jet spouting discharge valve 10 is seated at the drain port 14a, and the drain port 14a is closed. This stops the jet spouting from the jet spout port 2d.

[0080] In the present embodiment, the jet dead water level JWL is set to a level above the stepped portion 14e of the inner tank 14. Consequently, flush water stored at a comparatively high position in the second tank part 15 is used for spouting from the jet spout port 2d. Therefore, a comparatively high hydraulic head pressure acts on the flush water spouted from the jet spout port 2d, and the jet spouting can

be performed at a large flow rate with a strong water force. Furthermore, since the horizontal cross-sectional area A_2 above the stepped portion $14e$ of the inner tank 14 is large, a large amount of flush water can be acquired for jet spouting.

[0081] In a state in which the main valve body $20a$ of the ball tap 16 is opened, the flush water supplied from the water supply source 6 (water supply) is supplied to the hydraulic drive mechanism 18 via the ball tap 16 , and flows through the water supply pipe 32 (FIG. 5) connected to the cylinder $18a$ into the second tank part 15 . Here, in a state in which the jet spouting discharge valve 10 is opened, the flow rate of the flush water flowing out of the drain port $14a$ of the inner tank 14 is larger than the flow rate of the flush water flowing through the water supply port $32a$ of the water supply pipe 32 into the second tank part 15 , and hence the water level in the second tank part 15 decreases. Then, after the jet spouting is stopped at time t_3 in FIG. 8 (after the jet spouting discharge valve 10 is closed), the flush water flowing in from the water supply pipe 32 raises the water level in the second tank part 15 .

[0082] On the other hand, since the rim spouting discharge valve 8 is still in an opened state, the flush water in the first tank part 13 flows out of the first drain port $12a$, and the water level in the first tank part 13 continues to decrease. Then, at time t_4 in FIG. 8, when the water level in the flush water tank main body 12 decreases to a predetermined rim dead water level RWL (FIG. 3), the rim spouting discharge valve 8 is seated on the first drain port $12a$, and the rim spouting discharge valve 8 is closed. Consequently, the rim spouting from the rim spout port $2c$ is stopped.

[0083] In the present embodiment, the rim dead water level RWL is set to below the stepped portion $14e$ of the inner tank 14 . For this reason, even the flush water stored at a comparatively low position in the first tank part 13 is used in spouting from the rim spout port $2c$. Therefore, an exceptionally large hydraulic head pressure does not act on the flush water spouted from the rim spout port $2c$. However, the rim spout port $2c$, which opens above the pooled water surface of the bowl $2a$, is open to the atmosphere and can facilitate spouting even if the hydraulic head pressure of flush water is low, and the bowl $2a$ can be effectively flushed. Furthermore, according to the present embodiment, in the rim spouting, the flush water stored in the first tank part 13 is utilized from the initial water level above the stepped portion $14e$ to the rim dead water level RWL below the stepped portion $14e$, and hence a large amount of flush water can be acquired for the rim spouting.

[0084] Furthermore, even after the rim spouting discharge valve 8 is closed, the main valve body $20a$ of the ball tap 16 is maintained in an opened state, and the flush water supplied from the water supply source 6 (water supply) therefore flows into the second tank part 15 from the water supply port $32a$ of the water supply pipe 32 via the ball tap 16 and the hydraulic drive mechanism 18 .

[0085] Then, the water level in the second tank part 15 rises due to the inflow of flush water, and at time t_5 in FIG. 8 at which the second tank part 15 is full, the flush water begins to overflow from the inner tank 14 , and the flush water flows into the first tank part 13 .

[0086] As flush water flows into the first tank part 13 , the water level in the first tank part 13 begins to rise. Next, the water level in the first tank part 13 rises to a predetermined initial water level, and the float 22 of the ball tap 16 rises,

to close the pilot valve 26 (FIG. 4). Thus, when the pilot valve 26 is closed, the flush water flowing into the pressure chamber $20d$ from the bleed hole $20c$ provided in the main valve body $20a$ of the ball tap 16 cannot flow out, and the pressure in the pressure chamber $20d$ increases. Then, at time t_6 in FIG. 8, the main valve body $20a$ is pressed by the pressure in the pressure chamber $20d$ and seated on the valve seat $20b$, and the main valve body $20a$ is closed. This stops water supply from the water supply source 6 via the ball tap 16 to the hydraulic drive mechanism 18 and stops the supply of flush water into the second tank part 15 .

[0087] Upon stopping the water supply to the hydraulic drive mechanism 18 , the piston $18b$ (FIG. 5) in the cylinder $18a$ pushed up by the water supply is pushed down by the biasing force of the spring $18c$. The rod 28 attached to the piston $18b$ accordingly lowers. The rod 28 lowers to a predetermined position, and the rod 28 is recoupled to the valve shaft $10a$ of the jet spouting discharge valve 10 by the clutch mechanism 30 . As described above, toilet flushing is ended once, and the flush toilet apparatus 1 returns to the standby state for the toilet flushing.

[0088] The flush water tank device 4 of the embodiment of the present invention is configured so that the first tank part 13 and the second tank part 15 have different cross-sectional areas A_1 and A_2 depending on the height, and the ratio of the cross-sectional area of the first tank part 13 to that of the second tank part 15 at the first height H_1 (A_1/A_2 in FIG. 6) is different from the ratio of the cross-sectional area at the second height H_2 that is lower than the first height H_1 (A_1/A_2 in FIG. 6). For example, as in the above-described embodiment, when the second tank part 15 is configured so that cross-sectional area A_2 is large at the first height H_1 and small at the second height H_2 , a large amount of flush water can be spouted from the jet spout port $2d$ at a high hydraulic head pressure in the initial stage of flushing. That is, it is possible to distribute the flush water in the flush water tank main body 12 to the rim spout port $2c$ and the jet spout port $2d$ in the flush toilet main body 2 in accordance with the required amount of flush water and the water force. Consequently, the volume in the flush water tank main body 12 and the hydraulic head pressure of the flush water in the flush water tank main body 12 can be fully utilized for toilet flushing.

[0089] Furthermore, according to the flush water tank device 4 of the present embodiment, since the second tank part 15 is configured to have the cross-sectional area that is equal to or more than the cross-sectional area A_1 of the first tank part 13 at the first height H_1 , in the initial stage of flushing in which the water level in the second tank part 15 is high, the instantaneous flow rate of flush water spouted from the jet spout port $2d$ of the flush toilet main body can be increased, and flushing effect can be enhanced.

[0090] According to the flush water tank device 4 of the present embodiment, the second tank part 15 is configured to have the reduced cross-sectional area A_2 below the stepped portion $14e$, while the first tank part 13 is configured to have the enlarged cross-sectional area A_1 below the stepped portion $14e$. As a result, in the initial stage of flushing, the instantaneous flow rate from the jet spout port $2d$ can be increased, and concurrently, a sufficient volume can be acquired in the first tank part 13 at the height that is equal to or less than the height of the stepped portion $14e$, so that a sufficient amount of flush water can be allocated to the rim spout port $2c$.

[0091] According to the flush water tank device 4 of the present embodiment, the inner tank 14 is disposed in the flush water tank main body 12, the second tank part 15 is formed inside the inner tank 14, and the first tank part 13 is formed inside the flush water tank main body 12 and outside the inner tank 14. Therefore, the cross-sectional area of each of the first tank part 13 and the second tank part 15 at each height can be freely set, and the amount of flush water and the hydraulic head pressure can be freely set as required.

[0092] According to the flush water tank device 4 of the present embodiment, since the height of the stepped portion 14e in which the cross-sectional area A2 of the second tank part 15 is reduced is set to be lower than the jet dead water level JWL, the flow rate of flush water spouted from the jet spout port 2d can be maintained to be sufficiently high until the spouting is ended.

[0093] According to the flush water tank device 4 of the present embodiment, since the stepped portion 14e in which the cross-sectional area A1 of the first tank part 13 enlarges is provided at a position higher than the rim dead water level RWL, the flush water stored below the stepped portion 14e of the first tank part 13 can be spouted from the rim spout port 2c, and a sufficient amount of flush water spouted from the rim spout port 2c can be acquired.

[0094] Furthermore, according to the flush water tank device 4 of the present embodiment, since the first tank part 13 and the second tank part 15 are configured so that the ratio of the cross-sectional area (A1/A2) is reversed at the first height H1 and the second height H2, the instantaneous flow rate of flush water spouted from the jet spout port 2d can be increased. Concurrently, enough flush water spouted from the rim spout port 2c can be acquired.

[0095] As described above, the flush water tank device of the embodiment of the present invention and the flush toilet apparatus including the same have been described, but various changes can be made to the above-described embodiment. In the above-described embodiment, the first tank part and the second tank part are formed by disposing the inner tank in the flush water tank main body, and the first tank part and the second tank part may be formed by providing a partition wall in the flush water tank main body.

[0096] Furthermore, in the above-described embodiment, the horizontal cross-sectional area of the first tank part and the second tank part rapidly changes with the stepped portion provided in the inner tank, and the partition wall in the flush water tank main body or the inner tank may be configured so that the horizontal cross-sectional area of the first tank part and the second tank part changes gently depending on the height.

[0097] Furthermore, in the above-described embodiment, the ball tap is used as the water supply valve for switching the supply and stop of the flush water to be stored in the flush water tank main body and the inner tank, and the water supply valve other than the ball tap may be used. Furthermore, in the above-described embodiment, the jet spouting discharge valve is driven by the hydraulic drive mechanism, and the hydraulic drive mechanism may not be provided.

REFERENCE SIGNS LIST

- [0098] 1 flush toilet apparatus
- [0099] 2 flush toilet main body
- [0100] 2a bowl
- [0101] 2b discharge trap conduit
- [0102] 2c rim spout port

- [0103] 2d jet spout port
- [0104] 2e rim conduit
- [0105] 2f jet conduit
- [0106] 4 flush water tank device
- [0107] 4a lever handle
- [0108] 6 water supply source
- [0109] 8 rim spouting discharge valve (rim spouting switch mechanism)
- [0110] 8a bead chain
- [0111] 10 jet spouting discharge valve
- [0112] 10a valve shaft
- [0113] 12 flush water tank main body
- [0114] 12a first drain port
- [0115] 12b second drain port
- [0116] 13 first tank part
- [0117] 14 inner tank
- [0118] 14a drain port
- [0119] 14b drain port forming member
- [0120] 14c cutout
- [0121] 14d cutout
- [0122] 14e stepped portion
- [0123] 15 second tank part
- [0124] 16 ball tap (water supply valve)
- [0125] 16a inflow pipe
- [0126] 16b outflow pipe
- [0127] 18 hydraulic drive mechanism
- [0128] 18a cylinder
- [0129] 18b piston
- [0130] 18c spring
- [0131] 20 main body
- [0132] 20a main valve body
- [0133] 20b valve seat
- [0134] 20c bleed hole
- [0135] 20d pressure chamber
- [0136] 20e pressure passage
- [0137] 22 float
- [0138] 24 arm
- [0139] 24a support shaft
- [0140] 26 pilot valve
- [0141] 26a pilot valve port
- [0142] 28 rod
- [0143] 30 clutch mechanism
- [0144] 32 water supply pipe
- [0145] 32a water supply port

1. A flush water tank device that is attached to a flush toilet main body for use to supply flush water, comprising:

- a flush water tank main body including a first tank part and a second tank part that store flush water supplied to the flush toilet main body,
- a rim spouting switch mechanism that switches discharge and stop of flush water in the first tank part so that the flush water in the first tank part is spouted from a rim spout port of the flush toilet main body, and
- a jet spouting discharge valve that switches discharge and stop of flush water in the second tank part so that the flush water in the second tank part is spouted from a jet spout port of the flush toilet main body, wherein

the first tank part and the second tank part are configured to have different cross-sectional areas depending on a height, and a ratio of the cross-sectional area of the first tank part to the cross-sectional area of the second tank part at a first height is different from a ratio of the cross-sectional area at a second height that is lower than the first height.

2. The flush water tank device according to claim 1, wherein the second tank part is configured to have a cross-sectional area that is equal to or more than the cross-sectional area of the first tank part at the first height.

3. The flush water tank device according to claim 2, wherein the second tank part includes a stepped portion provided at a height between the first height and the second height, and the second tank part is configured to have a reduced cross-sectional area below the stepped portion, while the first tank part is configured to have an enlarged cross-sectional area below the stepped portion.

4. The flush water tank device according to claim 2, wherein an inner tank is disposed in the flush water tank main body, the second tank part is formed inside the inner tank, and the first tank part is formed inside the flush water tank main body and outside the inner tank.

5. The flush water tank device according to claim 3, wherein the jet spouting discharge valve is configured to discharge the flush water in the second tank part so that a water level in the second tank part lowers from a predetermined initial water level to a predetermined jet dead water

level, and the stepped portion is provided at a position that is lower than the jet dead water level.

6. The flush water tank device according to claim 3, wherein the rim spouting switch mechanism is configured to discharge the flush water in the first tank part so that the water level in the first tank part lowers from the predetermined initial water level to a predetermined rim dead water level, and the stepped portion is provided at a position that is higher than the rim dead water level.

7. The flush water tank device according to claim 1, wherein the first tank part and the second tank part are configured so that the ratio of the cross-sectional area is reversed at the first height and the second height.

8. A flush toilet apparatus that is flushed with flush water stored in a flush water tank, comprising:

a flush toilet main body including a bowl, a rim spout port, and a jet spout port, and

the flush water tank device according to claim 1, which supplies flush water to the rim spout port and the jet spout port of the flush toilet main body.

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